

Ordinary Differential Equations (ODE) — 30 Selected Questions

Topic 2.3: Ordinary Differential Equations (ODE) • 30 Selected PYQs

TOTAL QUESTIONS

30

TOTAL MARKS

66 M

TEST DURATION

90 Min

PAPER PATTERN

MCQ • MSQ • NAT

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Section A: MCQ	15	+1 / +2 Marks	$-1/3$ / $-2/3$ Marks	4 Options (1 Correct)
Section B: MSQ	5	+2 Marks	Nil (0)	4 Options (1+ Correct)
Section C: NAT	10	+1 / +2 Marks	Nil (0)	Real Decimal Value

IMPORTANT INSTRUCTIONS FOR CANDIDATES

- Authentic Examination PYQs:** This custom question paper comprises 30 verified official IIT JAM Mathematics (MA) questions on Ordinary Differential Equations.
- Section A (Multiple Choice Questions - MCQ):** Each question has four options, with **only one** correct choice. Wrong answers carry negative penalty: $1/3$ deduction for 1-mark questions, and $2/3$ deduction for 2-mark questions.
- Section B (Multiple Select Questions - MSQ):** Each question has four options, where **one or more than one** option(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial or negative marking.
- Section C (Numerical Answer Type - NAT):** Numerical answers are real numbers entered via keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
- Master Answer Key & Solutions:** Complete official answers, acceptance ranges, and evaluation rubrics are provided at the end of this document.
- Online Interactive Simulator:** Practice full tests with virtual calculator, timers, and instant scoring analytics at vaibhavgeometer.github.io.

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SECTION A: MULTIPLE CHOICE QUESTIONS (MCQ)

Question 1 (JAM 2015 • Q9 • ID: JAM_2015_Q9)

MCQ

+1 M

Neg: -0.33

Q.9 An integrating factor of the differential equation

$$\frac{dy}{dx} = \frac{2xy^2 + y}{x - 2y^3}$$

is

(A) $\frac{1}{y}$

(B) $\frac{1}{y^2}$

(C) y

(D) y^2

Question 2 (JAM 2021 • Q2 • ID: JAM_2021_Q2)

MCQ

+1 M

Neg: -0.33

Q. 2 Let $P : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function such that $P(x) > 0$ for all $x \in \mathbb{R}$. Let y be a twice differentiable function on $\mathbb{R}$ satisfying $y''(x) + P(x)y'(x) - y(x) = 0$ for all $x \in \mathbb{R}$. Suppose that there exist two real numbers a, b ($a < b$) such that $y(a) = y(b) = 0$. Then

(A) $y(x) = 0$ for all $x \in [a, b]$.

(B) $y(x) > 0$ for all $x \in (a, b)$.

(C) $y(x) < 0$ for all $x \in (a, b)$.

(D) $y(x)$ changes sign on (a, b) .

Question 3 (JAM 2023 • Q4 • ID: JAM_2023_Q4)

MCQ

+1 M

Neg: -0.33

Q. 4 Consider the initial value problem

$$\frac{dy}{dx} + \alpha y = 0,$$
$$y(0) = 1,$$

where $\alpha \in \mathbb{R}$. Then

- (A) there is an α such that $y(1) = 0$
- (B) there is a unique α such that $\lim_{x \rightarrow \infty} y(x) = 0$
- (C) there is NO α such that $y(2) = 1$
- (D) there is a unique α such that $y(1) = 2$

Question 4 (JAM 2024 • Q10 • ID: JAM_2024_Q10)

MCQ

+1 M

Neg: -0.33

Q.10 Let $g: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function. Which one of the following is the solution of the differential equation

$$\frac{d^2 y}{dx^2} + y = g(x) \text{ for } x \in \mathbb{R},$$

satisfying the conditions $y(0) = 0, y'(0) = 1$?

(A) $y(x) = \sin x - \int_0^x \sin(x-t) g(t) dt$

(B) $y(x) = \sin x + \int_0^x \sin(x-t) g(t) dt$

(C) $y(x) = \sin x - \int_0^x \cos(x-t) g(t) dt$

(D) $y(x) = \sin x + \int_0^x \cos(x-t) g(t) dt$

Question 5 (JAM 2026 • Q4 • ID: JAM_2026_Q4)

MCQ

+1 M

Neg: -0.33

Q.4	The general solution of the differential equation $x \sin\left(\frac{y}{x}\right) \frac{dy}{dx} = y \sin\left(\frac{y}{x}\right) + \frac{x}{2}, \quad x > 0$ is given by _____.
(A)	$\cos\left(\frac{x}{y}\right) + \log_e(x^2) = K$, where K is an arbitrary constant
(B)	$\cos\left(\frac{y}{x}\right) + \log_e(x^2) = K$, where K is an arbitrary constant
(C)	$\cos\left(\frac{x}{y}\right) + \log_e(\sqrt{x}) = K$, where K is an arbitrary constant
(D)	$\cos\left(\frac{y}{x}\right) + \log_e(\sqrt{x}) = K$, where K is an arbitrary constant

Question 6 (JAM 2016 • Q22 • ID: JAM_2016_Q22)

MCQ

+2 M

Neg: -0.67

Q.22 The general solution of the differential equation with constant coefficients

$$\frac{d^2y}{dx^2} + b \frac{dy}{dx} + cy = 0$$

approaches zero as $x \rightarrow \infty$, if

- (A) b is negative and c is positive
- (B) b is positive and c is negative
- (C) both b and c are positive
- (D) both b and c are negative

Question 7 (JAM 2024 • Q23 • ID: JAM_2024_Q23)

MCQ

+2 M

Neg: -0.67

Q.23 For $\alpha \in \mathbb{R}$, let $y_\alpha(x)$ be the solution of the differential equation

$$\frac{dy}{dx} + 2y = \frac{1}{1+x^2} \quad \text{for } x \in \mathbb{R}$$

satisfying $y(0) = \alpha$. Then, which one of the following is TRUE?

- (A) $\lim_{x \rightarrow \infty} y_\alpha(x) = 0$ for every $\alpha \in \mathbb{R}$
- (B) $\lim_{x \rightarrow \infty} y_\alpha(x) = 1$ for every $\alpha \in \mathbb{R}$
- (C) There exists an $\alpha \in \mathbb{R}$ such that $\lim_{x \rightarrow \infty} y_\alpha(x)$ exists but its value is different from 0 and 1
- (D) There is an $\alpha \in \mathbb{R}$ for which $\lim_{x \rightarrow \infty} y_\alpha(x)$ does not exist

Question 8 (JAM 2025 • Q11 • ID: JAM_2025_Q11)

MCQ

+2 M

Neg: -0.67

Q.11 Let $\mathcal{C}$ denote the family of curves described by $yx^2 = \lambda$, for $\lambda \in (0, \infty)$ and lying in the first quadrant of the xy plane. Let $\mathcal{O}$ denote the family of orthogonal trajectories of $\mathcal{C}$.

Which one of the following curves is a member of $\mathcal{O}$, and passes through the point $(2, 1)$?

- (A) $y = \frac{x^2}{4}, x > 0, y > 0$
- (B) $x^2 - 2y^2 = 2, x > 0, y > 0$
- (C) $x - y = 1, x > 0, y > 0$
- (D) $2x - y^2 = 3, x > 0, y > 0$

Question 9 (JAM 2025 • Q12 • ID: JAM_2025_Q12)

MCQ

+2 M

Neg: -0.67

Q.12 Let $\varphi : (0, \infty) \rightarrow \mathbb{R}$ be the solution of the differential equation

$$x \frac{dy}{dx} = (\ln y - \ln x)y,$$

satisfying $\varphi(1) = e^2$. Then, the value of $\varphi(2)$ is equal to

- (A) e^2
- (B) $2e^3$
- (C) $3e^2$
- (D) $6e^3$

Question 10 (JAM 2026 • Q16 • ID: JAM_2026_Q16)

MCQ

+2 M

Neg: -0.67

Q.16	The orthogonal trajectories of the family of curves $y = -3x - 3 + me^x \quad \text{for a real parameter } m$ are given by _____.
(A)	$x = \frac{y}{3} - \frac{1}{9} + k e^{-3y}$, k is a real parameter
(B)	$x = \frac{y}{3} - \frac{1}{9} + k e^{3y}$, k is a real parameter
(C)	$x = \frac{-y}{3} + \frac{1}{9} + k e^{-3y}$, k is a real parameter
(D)	$x = \frac{-y}{3} + \frac{1}{9} + k e^{3y}$, k is a real parameter

Question 11 (JAM 2026 • Q18 • ID: JAM_2026_Q18)

MCQ

+2 M

Neg: -0.67

Q.18

Let $\phi: \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \rightarrow \mathbb{R}$ be the solution of the differential equation

$$(\cos x) \frac{dy}{dx} - y = 2y^2(\sin x - 1) \cos x$$

satisfying $\phi(0) = \frac{1}{2}$. Then $\phi\left(\frac{\pi}{4}\right) = \underline{\hspace{2cm}}$.

(A) 2

(B) $\frac{3 + 2\sqrt{2}}{2}$ (C) $\frac{1}{2}$ (D) $\frac{1}{\sqrt{2}}$

Question 12 (JAM 2006 • Q10 • ID: JAM_2006_Q10)

MCQ

+6 M

Neg: -2.0

10. If $(c_1 + c_2 \ln x)/x$ is the general solution of the differential equation

$$x^2 \frac{d^2 y}{dx^2} + kx \frac{dy}{dx} + y = 0, \quad x > 0,$$

then k equals

(A) 3

(B) -3

(C) 2

(D) -1

Question 13 (JAM 2008 • Q9 • ID: JAM_2008_Q9)

MCQ

+6 M

Neg: -2.0

Q.9 Let $y_1(x)$ and $y_2(x)$ be linearly independent solutions of the differential equation $y'' + P(x)y' + Q(x)y = 0$, where $P(x)$ and $Q(x)$ are continuous functions on an interval I . Then $y_3(x) = a y_1(x) + b y_2(x)$ and $y_4(x) = c y_1(x) + d y_2(x)$ are linearly independent solutions of the given differential equation if

(A) $ad = bc$ (B) $ac = bd$ (C) $ad \neq bc$ (D) $ac \neq bd$

Question 14 (JAM 2011 • Q5 • ID: JAM_2011_Q5)

MCQ

+6 M

Neg: -2.0

Q.5 The solution $y(x)$ of the differential equation $\frac{d^2 y}{dx^2} + 4 \frac{dy}{dx} + 4y = 0$ satisfying the conditions $y(0) = 4$, $\frac{dy}{dx}(0) = 8$ is

(A) $4e^{2x}$ (B) $(16x + 4)e^{-2x}$ (C) $4e^{-2x} + 16x$ (D) $4e^{-2x} + 16xe^{2x}$

MA-1 / 40

Question 15 (JAM 2012 • Q6 • ID: JAM_2012_Q6)

MCQ

+6 M

Neg: -2.0

Q.6 An integrating factor for the differential equation $(2xy + 3x^2y + 6y^3)dx + (x^2 + 6y^2)dy = 0$ is

(A) x^3 (B) y^3 (C) e^{3x} (D) e^{3y}

SECTION B: MULTIPLE SELECT QUESTIONS (MSQ)

Question 16 (JAM 2016 • Q35 • ID: JAM_2016_Q35)

MSQ

+2 M

Neg: Nil (0)

Q.35 Consider the differential equation

$$\sin 2x \frac{dy}{dx} = 2y + 2 \cos x, \quad y\left(\frac{\pi}{4}\right) = 1 - \sqrt{2}.$$

Then which of the following statement(s) is(are) TRUE?

- (A) The solution is unbounded when $x \rightarrow 0$
- (B) The solution is unbounded when $x \rightarrow \frac{\pi}{2}$
- (C) The solution is bounded when $x \rightarrow 0$
- (D) The solution is bounded when $x \rightarrow \frac{\pi}{2}$

Question 17 (JAM 2017 • Q40 • ID: JAM_2017_Q40)

MSQ

+2 M

Neg: Nil (0)

Q.40 Let $k, \ell \in \mathbb{R}$ be such that every solution of

$$\frac{d^2 y}{dx^2} + 2k \frac{dy}{dx} + \ell y = 0$$

satisfies $\lim_{x \rightarrow \infty} y(x) = 0$. Then

- (A) $3k^2 + \ell < 0$ and $k > 0$
- (B) $k^2 + \ell > 0$ and $k < 0$
- (C) $k^2 - \ell \leq 0$ and $k > 0$
- (D) $k^2 - \ell > 0, k > 0$ and $\ell > 0$

Question 18 (JAM 2023 • Q33 • ID: JAM_2023_Q33)

MSQ

+2 M

Neg: Nil (0)

Q. 33 Let $y : (\sqrt{2/3}, \infty) \rightarrow \mathbb{R}$ be the solution of

$$(2x - y)y' + (2y - x) = 0,$$

$$y(1) = 3.$$

Then

- (A) $y(3) = 1$
- (B) $y(2) = 4 + \sqrt{10}$
- (C) y' is bounded on $(\sqrt{2/3}, 1)$
- (D) y' is bounded on $(1, \infty)$

Question 19 (JAM 2025 • Q33 • ID: JAM_2025_Q33)

MSQ

+2 M

Neg: Nil (0)

Q.33 Let $\varphi : (-1, \infty) \rightarrow (0, \infty)$ be the solution of the differential equation

$$\frac{dy}{dx} - 2ye^x = 2e^x\sqrt{y},$$

satisfying $\varphi(0) = 1$.

Then, which of the following is/are TRUE?

- (A) φ is an unbounded function
- (B) $\lim_{x \rightarrow \ln 2} \varphi(x) = (2e - 1)^2$
- (C) $\lim_{x \rightarrow \ln 2} \varphi(x) = \sqrt{2e - 1}$
- (D) φ is a strictly increasing function on the interval $(0, \infty)$

Question 20 (JAM 2026 • Q35 • ID: JAM_2026_Q35)

MSQ

+2 M

Neg: Nil (0)

Q.35	Consider the differential equation $(2y \cos x - xy \sin x) dx + 2x \cos x dy = 0 \quad \text{for } x \in \left(0, \frac{\pi}{4}\right).$ Which of the following is/are integrating factor(s) of the differential equation?
(A)	$\frac{1}{xy}$
(B)	xy
(C)	$\sec x$
(D)	$\sqrt{\sec x}$

SECTION C: NUMERICAL ANSWER TYPE (NAT)

Question 21 (JAM 2016 • Q47 • ID: JAM_2016_Q47)

NAT

+1 M

Neg: Nil (0)

Q.47 Let $y_1(x)$, $y_2(x)$ and $y_3(x)$ be linearly independent solutions of the differential equation

$$\frac{d^3 y}{dx^3} - 6 \frac{d^2 y}{dx^2} + 11 \frac{dy}{dx} - 6y = 0.$$

If the Wronskian $W(y_1, y_2, y_3)$ is of the form ke^{bx} for some constant k , then the value of b is _____

Question 22 (JAM 2020 • Q46 • ID: JAM_2020_Q46)

NAT

+1 M

Neg: Nil (0)

JAM 2020

MATHEMATICS - MA

Q. 46 Consider the differential equation

$$\frac{dy}{dx} + 10y = f(x), \quad x > 0,$$

where $f(x)$ is a continuous function such that $\lim_{x \rightarrow \infty} f(x) = 1$. Then the value of

$$\lim_{x \rightarrow \infty} y(x)$$

is _____.

Question 23 (JAM 2020 • Q48 • ID: JAM_2020_Q48)

NAT

+1 M

Neg: Nil (0)

Q. 48 Let $f(x, y) = 0$ be a solution of the homogeneous differential equation

$$(2x + 5y)dx - (x + 3y)dy = 0.$$

If $f(x + \alpha, y - 3) = 0$ is a solution of the differential equation

$$(2x + 5y - 1)dx + (2 - x - 3y)dy = 0,$$

then the value of α is _____.

Question 24 (JAM 2022 • Q43 • ID: JAM_2022_Q43)

NAT

+1 M

Neg: Nil (0)

Q.43

Let $y(x)$ be the solution of the differential equation

$$\frac{dy}{dx} + 3x^2y = x^2, \quad \text{for } x \in \mathbb{R},$$

satisfying the initial condition $y(0) = 4$.Then $\lim_{x \rightarrow \infty} y(x)$ is equal to _____. (Rounded off to two decimal places)

Question 25 (JAM 2024 • Q42 • ID: JAM_2024_Q42)

NAT

+1 M

Neg: Nil (0)

Q.42 Let $y: \mathbb{R} \rightarrow \mathbb{R}$ be the solution to the differential equation

$$\frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} + 5y = 1$$

satisfying $y(0) = 0$ and $y'(0) = 1$.

Then, $\lim_{x \rightarrow \infty} y(x)$ equals _____ (rounded off to two decimal places).

Question 26 (JAM 2017 • Q60 • ID: JAM_2017_Q60)

NAT

+2 M

Neg: Nil (0)

Q.60 If

$$y(x) = \int_{\sqrt{x}}^x \frac{e^t}{t} dt, \quad x > 0$$

Question 27 (JAM 2019 • Q60 • ID: JAM_2019_Q60)

NAT

+2 M

Neg: Nil (0)

Q.60 If $y(x)$ is the solution of the initial value problem

$$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = 0, \quad y(0) = 2, \quad \frac{dy}{dx}(0) = 0,$$

then $y(\ln 2)$ is (round off to 2 decimal places) equal to _____

END OF THE QUESTION PAPER

Question 28 (JAM 2020 • Q57 • ID: JAM_2020_Q57)

NAT

+2 M

Neg: Nil (0)

JAM 2020

MATHEMATICS - MA

Q. 57 If $x^2 + xy^2 = c$, where $c \in \mathbb{R}$, is the general solution of the exact differential equation

$$M(x, y) dx + 2xy dy = 0,$$

then $M(1, 1)$ is _____.

[9 2 7 1]

Question 29 (JAM 2023 • Q59 • ID: JAM_2023_Q59)

NAT

+2 M

Neg: Nil (0)

Q. 59 Let S be the set of all real numbers α such that the solution y of the initial value problem

$$\begin{aligned}\frac{dy}{dx} &= y(2 - y), \\ y(0) &= \alpha,\end{aligned}$$

exists on $[0, \infty)$. Then the minimum of the set S is equal to _____.
(rounded off to two decimal places)

Question 30 (JAM 2024 • Q51 • ID: JAM_2024_Q51)

NAT

+2 M

Neg: Nil (0)

Section C: Q.51 – Q.60 Carry TWO marks each.

Q.51 For $\alpha \in (-2\pi, 0)$, consider the differential equation

$$x^2 \frac{d^2 y}{dx^2} + \alpha x \frac{dy}{dx} + y = 0 \quad \text{for } x > 0.$$

Let D be the set of all $\alpha \in (-2\pi, 0)$ for which all corresponding real solutions to the above differential equation approach zero as $x \rightarrow 0^+$. Then, the number of elements in $D \cap \mathbb{Z}$ equals _____

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

Ordinary Differential Equations (ODE) — 30 Selected Questions • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	MCQ	+1	B
2	MCQ	+1	A
3	MCQ	+1	D
4	MCQ	+1	B
5	MCQ	+1	D
6	MCQ	+2	C
7	MCQ	+2	A
8	MCQ	+2	B
9	MCQ	+2	B
10	MCQ	+2	C

Q. No.	Type	Marks	Official Key
11	MCQ	+2	D
12	MCQ	+6	A
13	MCQ	+6	C
14	MCQ	+6	B
15	MCQ	+6	C
16	MSQ	+2	C, D
17	MSQ	+2	C, D
18	MSQ	+2	B, D
19	MSQ	+2	A, B, D
20	MSQ	+2	B, D

Q. No.	Type	Marks	Official Key
21	NAT	+1	6.0 to 6.0
22	NAT	+1	0.1 to 0.1
23	NAT	+1	7 to 7
24	NAT	+1	0.32 to 0.34
25	NAT	+1	0.19 to 0.21
26	NAT	+2	1.34 to 1.36
27	NAT	+2	1.12 to 1.25
28	NAT	+2	3 to 3
29	NAT	+2	−0.01 to 0.01
30	NAT	+2	6 to 6

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt ($-1/3$ for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., $[a, b]$) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.